

RESEARCH ARTICLE

The effect of reminiscence therapy on cognitive functions, depression, and quality of life in Alzheimer patients: Randomized controlled trial

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Background: Alzheimer is a disease leading to various neuropsychiatric behavioral disorders, and the most common symptom observed during the prognosis of Alzheimer's disease is dysmnnesia. The aim of the present study is to investigate the effect of reminiscence therapy on cognitive functions, depression, and quality of life in Alzheimer's patients.

Methods: The present study was a randomized controlled single blind study with two groups, which was designed in the experimental pretest-posttest pattern in the city of Konya, Turkey. It was decided to include a total of 60 elderly individuals. Information form that evaluated socio-demographic characteristics and disease history of individuals in intervention and control groups and was developed by the researcher. Standardized Mini-Mental State Examination (SMMSE), Cornell Scale for Depression in Dementia, and the Quality of Life in Alzheimer's Disease (QOL-AD) Scale were used to collect the data. Reminiscence therapy was applied once a week and lasted for 8 weeks. Every session took 60 minutes. Groups consisted of six people.

Results: Mini-Mental Test, depression, and quality of life mean scores of the elderly in intervention group before reminiscence therapy program increased after the administration, and the difference was statistically significant ($P < 0.05$). In the inter-group comparison, a significant difference was found between elderly individuals' posttest Mini-Mental Test, depression, and quality of life mean scores ($P < 0.05$).

Conclusions: Our results suggest that regular reminiscence therapy should be considered for inclusion as routine care for the improvement of cognitive functions, depressive symptoms, and quality of life in elderly people with Alzheimer.

KEYWORDS

Alzheimer, cognitive functions, depression, quality of life, reminiscence therapy

1 | INTRODUCTION

Alzheimer's disease and mild cognitive impairment lead to various neuropsychiatric behavioral disorders and impair daily activities as well as cognitive decline. Amnesic mild cognitive impairment and Alzheimer's disease are characterized by decreased memory, which is assessed by the ability to learn and remember new information. Patients may experience cognitive and functional losses during course of the

disease and have problems in simple daily activities such as bathing, dressing, eating, and excretion.^{1,2} This condition may adversely influence patients' quality of life. It is important for Alzheimer's patients to meet their basic needs such as learning, thinking, feeling, capacity recognition, environmental recognition, and experiencing love and being a beloved person.³

Psychosocial intervention is a necessary process in order to improve the quality of life in Alzheimer's disease (QOL-AD).

Intervention methods applied to Alzheimer's patients include cognitive, behavioral, affective, and psychosocial interventions. Cognitive practices include cognitive training, cognitive stimulation, daily activities, and reality orientation therapy; behavioral practices include multifactorial behavioral interventions; affective practices include aromatherapy, massage, therapeutic touch, and music; and psychosocial practices include reminiscence, confirmatory therapy, and exercise therapy.^{4,5} In the meta-analysis by Sitzler, Twamley, and Jeste, it was stated that the effect size of cognitive training interventions was medium, patients gained cognitive and functional benefits, and cognitive training had positive effects on learning, memory, functionality, and daily activities.⁶ In another study on nonpharmacological implementations applied in nursing, houses in England were evaluated using a form developed by the researchers; therapies (64%) were commonly used among nonpharmacological practices applied for Alzheimer's and other types of dementia.⁷

Reminiscence group therapy is defined as vocal or silent recall of life events performed in the presence of a group. Reminiscence group therapy involves the discussion of past activities, events, and experiences, usually with the aid of tangible prompts.⁸ Reminiscence group therapy aims to share positive experiences in group environment and thus to make elderly individuals to feel themselves stronger, valuable, and self-confident.⁹ Tadaka and Kanagawa studied the effects of reminiscence group therapy on adapting to daily life and concluded that reminiscence therapy had a positive influence on Alzheimer's patients.¹⁰ Various studies reveal that reminiscence therapy has positive outcomes for elderly in terms of variables such as mood and some cognitive abilities, quality of life, well-being, communication, ego-integrity, and depression.¹¹⁻¹³

Since depressive symptoms and depressed mood are common among dementia patients, an association between poor affect and cognitive decline has been identified in several studies.¹⁴⁻¹⁶ In a meta-analysis by Chi and colleagues, the prevalence of depression was found as 12.7% (CI, 8.8-17.8) and 42% (CI, 38-45) according to the DSM criteria for major depression and the specific criteria for dementia, respectively.¹⁷ Bierman et al studied the effects of depression on cognition in later life and concluded that symptoms of depression always had a negative influence on cognitive functioning.¹⁴ According to the results from another study, the researcher stated that depressed mood was associated with a worse subsequent cognitive decline in Alzheimer's disease.¹⁸ There is some evidence that reminiscence therapy assists in the reduction of symptoms of depression in older people.^{8,13,19,20} Many studies have demonstrated that reminiscence therapy can improve psychosocial well-being²¹ and affect balance,²² reduce depression,²³ increase self-esteem and life satisfaction, and decrease depressive symptoms.²⁴ In addition, reminiscence therapy has been shown to increase life satisfaction in elderly.²⁵

Studies on reminiscence and its effects on depression vary in methodology or settings, making it difficult to compare the results and to determine the feasibility of reminiscence as an effective intervention. Because of the cultural differences, themes and materials that used in reminiscence therapy differ from country to country. In Turkey, nursing home services have largely focused toward health problems and the physical needs of the elderly. There is a considerable gap between the number of elderly people who reside in the nursing

Key points

- Reminiscence therapy has increased the cognitive functions of Alzheimer's patients.
- Reminiscence therapy has improved the levels of depression in Alzheimer's patients.
- Reminiscence therapy has increased the quality of life of Alzheimer's patients.
- Reminiscence therapy is an applicable therapy in Alzheimer patients.

home and the number of nurses working in these institutions. Using organized activities, nurses deal with the challenges of meeting the social and psychological needs of the elderly people who reside in nursing homes. For all these reasons, reminiscence therapy cannot be implemented in the nursing homes in Turkey. This therapy can only be implemented on the patients included in studies.

Implementing the reminiscence therapy in people with Alzheimer disease, whose quality of life and cognitive functions are negatively influenced, has been intended to be included within routine practices of elderly care centers, because it is an efficient and cost-effective method. Based on these goals, the objective of this study is to investigate the effect of reminiscence therapy on cognitive functions, depression, and quality of life in people with Alzheimer's disease.

2 | METHODS

The present study was a randomized controlled study with two groups and was designed in the experimental pretest-posttest pattern.

2.1 | Participants

The selection of participants and administration of the therapy were carried out in a nursing home affiliated with Konya Provincial Directorate of Family and Social Policies between September 2016 and October 2016.

Inclusion criteria included being at least primary school graduate, being diagnosed with typical Alzheimer's disease according to International Working Group-2 (IWG-2) diagnostic criteria, having a score between 13 and 24 points from Standardized Mini-Mental State Examination (SMMSE), continuing their regular treatment with acetylcholinesterase inhibitors, and having a score of 8 or higher points from Cornell Scale for Depression in Dementia (CSDD). Patients who participated in an accompanying similar program, those did not participate in at least two sessions of reminiscence therapy, illiterate patients, those diagnosed with other types of dementia, and had a SMMSE score lower than 13/30 and higher than 24/30 were excluded from the study.

2.2 | Sample size

Population of the study consisted of individuals diagnosed with Alzheimer's and who were residing in a nursing home affiliated with

Konya Provincial Directorate of Family and Social Policies. In order to calculate the sample size, effect size (Cohen *d*) was assumed as 0.66, power as 0.80, and type 1 error as 0.05, and the sample size was calculated as total 60 persons for two groups, and it was decided to include a total of 60 patients with 30 patients being in the intervention group and 30 patients in the control group. Power score of the total scores of these three scales (SMMSE, CSDD and QOL-AD) was calculated in order to determine effect size of the intervention. The lowest power was found in SMMSE by 0.88.

2.3 | Randomization

Seven-one patients who met the inclusion and exclusion criteria were determined from the files of 83 patients who were residing in the nursing home and diagnosed with Alzheimer's disease. The patients were numbered with 1 to 71 envelopes. Numbers of 71 patients were enveloped by a person who was not included in the study, placing the number of patients 1 coming from randomization into the envelope of patient 1 (Figure 1). Seven-one persons given numbers were assigned as 30 in the intervention group and 30 in the control group by envelope method. None of these persons rejected the study or left during the study.

2.4 | Data collection technique and tools

Data were collected using an information form that was developed by the researcher and evaluated socio-demographic characteristics and disease history of individuals in the intervention and the control groups, SMMSE, CSDD, and the QOL-AD Scale. Pretest and posttest data of the intervention group were collected by the researcher through face-to-face interview method. Data were collected by a single researcher, and the intervention was performed again by the same person.

2.5 | Standardized Mini-Mental State Examination

This test was developed by Folstein et al and can easily be applied. It provides information about the stages of the cognitive disorder. The Turkish validity and reliability study of the SMMSE were conducted by Güngen et al.²⁶ The test consists of orientation, recording, attention, calculation, reminding, language tests, and structuring sections. Each question in the test gets 1 point. The lowest and highest scores to be obtained from the scale are 0 and 30, respectively. Scores between 0 and 12 indicate a "severe cognitive disorder," those between 13 and 22 indicate "midstage cognitive disorder," and those between 23 and 24 indicate "early-stage cognitive disorder." Scores between 25 and 30 suggest "no cognitive disorder." At the same time, dementia stage was determined with SMMSE. The correlation ($r = 0.99$) and kappa (0.92) values between the total scores obtained from the scale were found to be high.²⁶ It took about 5 to 10 minutes to administer the test.

2.6 | Cornell Scale for Depression in Dementia

Cornell Scale for Depression in Dementia was developed by Alexopoulos, Abrams, Young, and Shamonian.²⁷ The Turkish validity and reliability study of the scale was conducted by Amuk et al,²⁸ and the Cronbach α value of the scale is 0.864. Cornell Scale for Depression in Dementia is a scale that is completed by the researcher according to the information obtained from the demented patients and their caregivers and enables to define the patient's depressive signs. The scale consists of 19 items in five subscales evaluating mood-related findings, physical findings, behavioral changes, cyclic functions, and cognitive changes. Each item is scored between 0 and 2. A score of 8 or over on the scale indicates depression.²⁸ It took about 5 to 10 minutes to administer the scale.

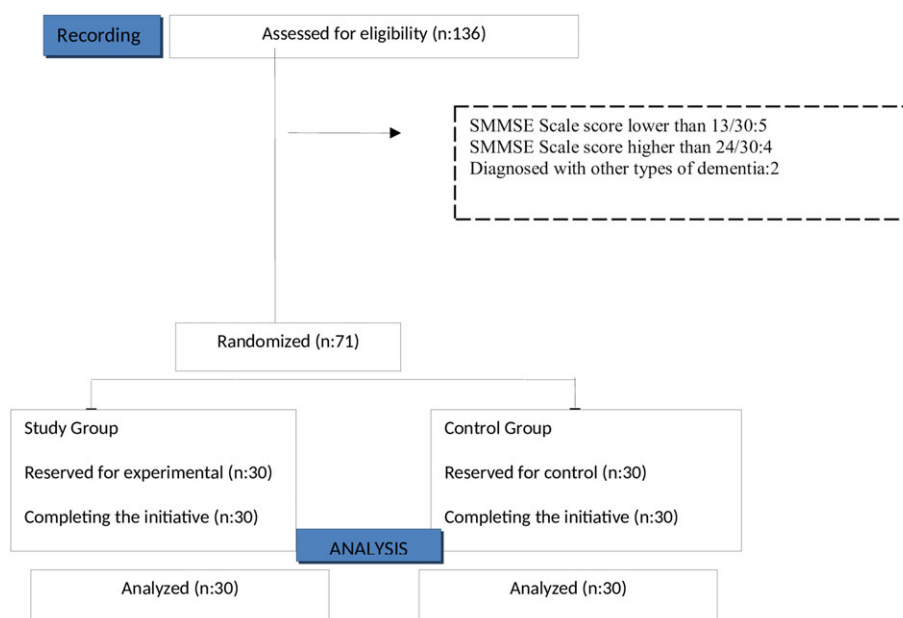


FIGURE 1 Flow diagram CONSORT. References: http://www.consortstatement.org/Media/Default/Downloads/Translations/Turkish_tr/Turkish%20CONSORT%20Flow%20Diagram.pdf. SMMSE, Standardized Mini-Mental State Examination [Colour figure can be viewed at wileyonlinelibrary.com]

2.7 | QOL-AD Scale

QOL-AD was developed by Logsdon, Gibbons, and Teri (2002).²⁹ This scale was adapted into Turkish by Akpınar and Küçükçüçlü.³⁰ This scale consists of having 13 items providing information about patients' quality of life. Each item in QOL-AD is scored on a point scale ranging from 1 to 4 (1 = poor, 4 = great). Scoring is based on the simple aggregation of the points obtained from all items. Scores obtained from the patients and caregivers can be separately calculated or assessed as a single score. For every marked item, point 1 indicates poor, point 2 moderate, point 3 good, and point 4 great. Scores to be obtained from the scale range between 13 and 52. QOL-AD form's internal consistency reliability coefficient (Cronbach's alpha coefficient) of the patients and intragroup correlation coefficient were found as 0.84 and 0.79, respectively. Higher scores from the scale indicate better quality of life.³⁰

2.8 | Intervention

Reminiscence therapy was applied once a week and lasted for 8 weeks. Every session took 60 minutes. Individuals in the intervention group were divided in five groups. Each group consisted of six people. Sessions included the titles of the first meeting, childhood experiences, festivals, memorable travelled places, favorite foods, important historical terms, achievements, and music of the term. The patients were encouraged to remember their important experiences, positive experiences, and achievements in the past by reminiscence therapy, and they were allowed to interactively share them verbally with group members. Group leader provided a supportive approach to ensure interaction and share between the participants and to make them feeling stronger, valuable, and self-confident. Materials triggering the memory, such as photographs, household goods, other reminiscent of the past, music of old times, and foods, were used.

2.9 | Statistical analysis

Statistical analysis was carried out using SPSS 18.0 (IBM, Armonk, USA). While continuous data are expressed as median (interquartile range) values, categorical data are given as numbers and percentages. The researchers used descriptive statistical tools including numbers, percentages, means, and standard deviation. One-way ANOVA test and *t* test were used to compare pretest-posttest mean scores of the intervention and the control groups. The Wilcoxon signed rank test was used to assess the pretest and posttest data of the intervention group and the control group within themselves.

2.10 | Ethical consideration

Ethical approval was obtained from Selcuk University, Faculty of Medicine, Ethical Committee of Clinical Studies in order to conduct the study. All participants in the intervention and the control groups were informed about the title, objective, duration, and type of the study, and the consent form was read aloud. Thus, they were ensured to understand the objective and content of the study. Written consent was obtained from those who agreed to participate in the study. Data collection and application phases were initiated after the consents were obtained.

3 | RESULTS

Comparison of the data obtained from the participants regarding their socio-demographic characteristics and disease properties showed that there was no statistically significant difference between the intervention group and the control group. Table 1 shows the results related to the socio-demographic characteristics and disease properties of the participants.

The Mini-Mental State Examination (MMSE) was administered before and after the "reminiscence therapy," and the difference between the total scale scores was examined. The Wilcoxon test was used to test the significance between the intervention group and the control group. It was found that Mini-Mental Test mean score of the elderly in the intervention group before reminiscence therapy program increased after the administration and the difference was statistically significant ($P < 0.05$). The one-way ANOVA test was used for the intergroup comparison because it was not normally distributed. No difference was found between the MMSE pretest mean scores of the intervention group and the control group ($P > 0.05$), whereas a significant difference was found between their posttest mean scores ($P < 0.05$) (Table 2).

The CSDD was administered before and after the "reminiscence therapy," and the difference between the total scale scores was examined. The Wilcoxon test was used to test the significance between the intervention group and the control group. Mean depression score of the elderly in the intervention group before the reminiscence therapy program increased after the administration, and the difference was

TABLE 1 The descriptive characteristics of the elderly in the intervention group and the control group

Socio-demographic Characteristics	Experimental Group (n = 30)		Control group (n = 30)		P
	n	%	n	%	
Gender					0.60
Female	18	60.0	16	53.4	
Male	12	40.0	14	46.6	
Educational level					0.19
Literate	5	16.6	6	20.0	
Primary education	12	40.0	11	36.6	
High school and above	13	43.4	13	43.4	
Evaluation of dementia					0.79
First stage	19	63.3	18	0.60	
Middle stage	11	36.7	12	0.40	

TABLE 2 The distribution of the Standardized Mini-Mental State Examination scores of the intervention group and the control group

SMMSE	Intervention Group (n = 30) M ± SD	Control Group (n = 30) M ± SD	F**	P
Baseline	18.26 ± 3.69	18.83 ± 3.00	1.457	0.687
Postintervention	21.08 ± 1.94	18.26 ± 1.74	0.761	0.001
Z value*	−2.745	−1.065		
P	0.001	0.352		

*Wilcoxon analysis, SD = 2.

**F: One-way ANOVA.

statistically significant ($P < 0.05$). The one-way ANOVA test was used for the intergroup comparison because it was not normally distributed. No difference was found between depression scale pretest mean scores of the intervention group and the control group ($P > 0.05$), whereas a significant difference was found between their posttest mean scores ($P < 0.05$) (Table 3).

Quality of Life Scale was administered before and after the "remembrance therapy," and the difference between the total scale scores was examined. The Wilcoxon test was used to test the significance between the intervention group and the control group. Mean quality of life score of the elderly in the intervention group before remembrance therapy program increased after the application, and the difference was statistically significant ($P < 0.05$). The one-way ANOVA test was used for the intergroup comparison, because it was not normally distributed. No difference was found between Quality of Life Scales pretest mean scores of the intervention group and the control group ($P > 0.05$), whereas a significant difference was found between their posttest mean scores ($P < 0.05$) (Table 4).

4 | DISCUSSION

The present study was conducted with the elderly individuals with Alzheimer's, who were residing in a nursing home; the effect of remembrance therapy on cognitive functions, depression levels, and quality of life in elderly individuals was evaluated. Results obtained from the study support the opinion that remembrance therapy has a positive effect on cognitive functions, depression levels, and quality of life in elderly.

TABLE 3 The distribution of the Cornell Scale for Depression in Dementia scores of the intervention group and the control group

CSDD	Intervention Group (n = 30) M ± SD	Control Group (n = 30) M ± SD	F**	P
Baseline	10.66 ± 3.61	11.43 ± 3.78	0.223	0.363
Post intervention	8.86 ± 1.56	11.76 ± 1.69	0.637	0.001
Z value*	1.657	2.086		
P	0.020	0.570		

Abbreviation: CSDD, Cornell Scale for Depression in Dementia.

*Wilcoxon analysis, SD = 2.

**F: One-way ANOVA.

TABLE 4 The distribution of the Quality of Life Scale scores of the intervention group and the control group

QOL-AD	Intervention Group (n = 30) M ± SD	Control Group (n = 30) M ± SD	F**	P
Baseline	18.05 ± 3.17	17.25 ± 3.78	0.346	0.74
Post intervention	20.22 ± 3.61	17.45 ± 3.41	0.748	0.001
Z value*	-7.778	-0.126		
P	0.001	0.120		

Abbreviation: QOL-AD, Quality of Life in Alzheimer's Disease.

*Wilcoxon analysis, SD = 2.

**F: One-way ANOVA.

Our results are consistent with the findings of previous studies, which have reported benefits of remembrance therapy in persons with Alzheimer's disease. Results of the remembrance therapy for individuals with Alzheimer's disease indicated that a significant difference was found between preintervention and postintervention cognitive functions of Alzheimer's patients. It is thought that the cognitive functions score was increased through the remembrance therapy and that the achievement obtained through the memory recall exercise had the positive effect. This result was consistent with findings of the previous studies. In a study by Wu and Koo, significant improvement was observed in cognitive impairment in elderly with dementia after the remembrance therapy.²⁵ Another study revealed that remembrance therapy had mild to moderate effect on cognitive functions.³¹ In their meta-analysis evaluating the efficiency of remembrance therapy, Huang et al indicated that remembrance therapy had a small-size effect on cognitive functions and a moderate-size effect on depressive symptoms.³¹ In accordance with the results of other studies and those obtained from the present study, practices supporting remembrance in elderly people were revealed to have a positive effect on cognitive process. For the cognitive function level, the experimental group also had higher scores than the control group, so that remembrance therapy needs to be continuously applied for the improvement of the cognitive function in people with Alzheimer's disease.

In the present study, at the end of a remembrance therapy, a significant decrease was observed in the scores of the depressive signs in people with Alzheimer disease intervention compared with controls. In a previous study, remembrance therapy for Alzheimer's patients was recommended as a highly effective treatment for depression and was also shown to improve quality of life and protect mental health.³² A meta-analysis revealed that remembrance therapy is effective in improving depressive symptoms in elderly people with dementia.³⁰ In a randomized controlled study by Meléndez Moral et al, it was reported that remembrance therapy significantly reduced depressive symptoms of elderly individuals.³³ Therefore, remembrance therapy has a positive effect not only on cognitive function but also on emotional functions in the people with Alzheimer's disease. However, Chao et al found no effect of remembrance therapy on depression.³² The results indicate that remembrance might be an effective therapy for depressive symptoms in Alzheimer's patients. Remembrance therapy and its effects on depression vary in methodology or settings, making it difficult to compare the results and to determine the feasibility of remembrance as an effective intervention.

In addition, the difference in quality of life scores between the experimental group, which received remembrance therapy, and the control group, which did not receive therapy, was examined. The result indicated that the experimental group had higher scores in quality of life than the control group. This result was consistent with the results of the previous studies.^{25,33} O'shea et al reported that remembrance had a positive effect on quality of life, as measured in the short-term over an 18- to 22-week period, significantly when examined on a per-protocol basis.³⁴ Life satisfaction can be viewed as a cognitive aspect of perceived quality of life.³³ It is concerned with daily issues that one encounters with emotions such as happiness, enjoyment, and contentment, and it is often perceived in comparison with one's past and one's friends.³⁵

The fact that reminiscence therapy improved the quality of life in a short time is thought to be associated with happiness of people with Alzheimer's disease resulting from remembering their past experiences, social support perception originating from getting together with peers, sharing similar experience, elevated self-confidence, sense of belonging to a group, and decreased depressive symptoms. In this situation, this study is significant in that it tested and verified the effect of reminiscence therapy as a nursing intervention method for improving cognitive function quality of life and decreasing depressive signs in people with Alzheimer's disease. Additionally, this study is significant in that it is a front experimental study verifying the effects of reminiscence therapy for people with Alzheimer's disease.

5 | LIMITATIONS

Because data of this study were collected using a convenience sample approach, generalizability of the study results is limited. The present study includes sample limitation because it was conducted in people with Alzheimer disease who were residing in a nursing home. Also, since premeasurement and postmeasurement in both groups were made using the memory practice method, results might be affected even in the least scale. As another limitation of this study, the control group did not administered any intervention. Finally, postintervention follow-up could not be made in the experiment and the control groups.

6 | CONCLUSIONS

Our results suggest that regular reminiscence therapy should be considered to be included as a routine care for the improvement of cognitive functions, depressive symptoms, and quality of life in elderly people with Alzheimer's disease. For further evaluation, a replication of this study using larger samples drawn from multiple communities is recommended to confirm the generalizability of its results. Further experimental studies for randomized trial and maintenance effects are also needed. It is recommended for nurses and all other health care professionals working with elderly people to routines "reminiscence therapy," which would be planned in accordance with their needs and physiological characteristics in order to protect them from chronic diseases and to ensure them to maintain a more qualified life.

CONFLICT OF INTEREST

No conflict of interest has been declared by the authors.

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